

The Impact of Model Quantization on Translation Quality Across Varying Sentence Difficulties

Mees Lindeman
University of Amsterdam
mees.lindeman@student.uva.nl

Ruben Figge
University of Amsterdam
ruben.figge@student.uva.nl

Melle van der Brugge
University of Amsterdam
melle.vanderbrugge@student.uva.nl

Tijs Wiegman
University of Amsterdam
tijs.wiegman@student.uva.nl

Matthias Hagenauer
University of Amsterdam
matthias.hagenauer@student.uva.nl

Abstract

This study investigates how model quantization impacts the translation quality of large language models (LLMs) across sentences of varying difficulty. Using TowerMistral 7B, a translation-specialized model, and Gemma-3, a general-purpose instruction-tuned model, we evaluate translation quality on a subset of the WMT24++ benchmark. Sentence difficulty is estimated automatically using a model trained on Error Span Annotation (ESA) data, enabling stratified analysis across easy, medium, and hard segments. Results show that both models maintain high semantic fidelity under moderate quantization (8- and 4-bit), but diverge significantly at 2-bit precision: Gemma-3 exhibits substantial degradation, while TowerMistral remains largely unaffected. These findings suggest that architectural specialization for multilingual translation enhances robustness to extreme compression, highlighting important trade-offs between model efficiency and translation quality. Code available on [GitHub](#).

1 Introduction

Large Language Models (LLMs) have shown remarkable versatility across natural language processing (NLP) tasks, from question answering and summarization to dialogue and machine translation (MT) (OpenAI, 2024; Touvron et al., 2023; Zhao et al., 2025). When appropriately prompted, an LLM can be used as a Neural Machine Translation (NMT) system. However, their large memory and computational demands hinder real-time or resource-limited deployment (Ataman et al., 2025).

To improve the efficiency and memory footprint of large models, compression methods like pruning, knowledge distillation, and quantization have been explored (Zhu et al., 2024). Quantization, which represents model parameters with lower precision formats (e.g., 16 $\rightarrow$ 8 or 4 bits), reduces model size and inference cost but can come with a tradeoff of degraded downstream performance.

This work investigates how quantization affects LLM-based MT across sentences of varying translation difficulty.

2 Related Work

Early Transformer quantization research showed that 8-bit integer representations could replace 32-bit floating-point computations with minimal performance degradation, yielding significant speedups (Bhandare et al., 2019). Pushing further, Aji and Heafield (2019) explored 4-bit quantization and found that logarithmic quantization preserves quality more effectively than fixed-point approaches. Mixed-precision strategies in transformers enable sub-3-bit quantization with under 0.5 BLEU point drop Chung et al. (2020). Together, these studies demonstrate that aggressive quantization can maintain most NMT performance while greatly improving efficiency.

However, recent work raises the concern that compression may unevenly affect model behavior. Mohammadshahi et al. (2022) observed that while overall metrics remained stable while performance on underrepresented languages or features often deteriorates significantly. Furthermore, compression can exacerbate existing gender and semantic biases, underscoring the necessity of evaluating compressed models beyond aggregated metrics.

Another line of research has examined the relationship between sentence-level difficulty and translation quality. Wan et al. (2022) revealed that short texts present unique challenges, such as over-translation and mistranslation, due to imbalanced training distributions and limited contextual cues. Similarly, Li et al. (2020) proposed techniques for explicitly compressing sentences to improve the representation of backbone information and thereby improve translation performance on structurally complex sentences. Agrawal and Carpuat (2019) further explored controlling text complexity

in neural machine translation (NMT), demonstrating that models can be guided to generate translations at different reading levels. This underscores the role of sentence difficulty in shaping translation outputs.

Taken together, prior work demonstrates that quantization can drastically reduce model size and computational costs with limited impact on global quality metrics. However, it may also introduce subtle degradations revealed when translation difficulty varies across inputs.

3 Dataset

For evaluation, we used a subset of the WMT24++ benchmark dataset (Deutsch et al., 2025), which uses English as the source language and covers 55 target languages and dialects. It contains 998 English source sentences taken from four different domains: literary, news, social, and speech. For our subset, we selected $en \rightarrow \{de, es, nl, zh\}$ (German, Spanish, Dutch, Chinese respectively).

To address the absence of difficulty scores in this dataset, we used a model provided by Proietti et al. (2025) to estimate difficulty scores for each source sentence. This method is trained on annotated datasets containing Error Span Annotation (ESA) scores, a metric based on human annotators identifying error spans and assigning severity labels (e.g., minor, major) to each, then combining these into an overall difficulty score (Kocmi et al., 2024b). We normalized the difficulty scores from the model to fall between 0 (Hard) and 1 (Easy). The difficulty score distribution is plot in Figure 1.

To analyse and compare performance for different difficulty scores, we split the sentences into three difficulty labels: ‘Easy’, ‘Medium’ & ‘Hard’. Note that ‘difficulty’ is a relative description and depends on many factors, including the model. Thus, the difficulty score cannot be used to decide objective difficulty, only relative, and bin borders can be

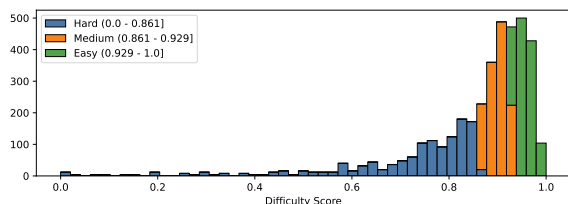


Figure 1: Distribution of estimated difficulty scores across all source sentences. Colours difficulty label. The ‘bleeding’ colors is due the bars not distinguishing the bin edges.

arbitrary as long as ordering is preserved. We opted for equal-sized bin populations to avoid sparsely populated bins, which can be more sensitive to outliers.

3.1 Bin statistics analysis

To analyse the bins, we computed the sentence length distribution per difficulty bin. ‘Easy’ sentences are mostly short, whereas ‘Hard’ sentences are both long and short, but longer on average, see Figure 2 (Top).

Using the wordfreq library, we also computed a rarity score $-\log(f + \epsilon)$ per word, where f is the word frequency and $\epsilon = 10^{-9}$ is used to avoid zero-errors. The higher the score, the higher the rarity. Taking the mean rarity score per sentence and plotting per bin, we find sentences in ‘Easy’ to have both the lowest and the highest rarity score, see Figure 2 (Bottom). We think that lower scores, i.e., common words, are easy to translate, and higher scores relate to special cases like URLs and emojis, where to translate is to change nothing.

Finally, we look qualitatively at $en \rightarrow nl$ sentences. We sort on difficulty score and find the easy sentences to be short with strong alignment and harder sentences to be long and consisting of many (sub)sentences. We also looked at the shortest sentences in the ‘Hard’ bin and found them to contain words that aren’t really words (like usernames), sentences without proper grammar and words with no translation.

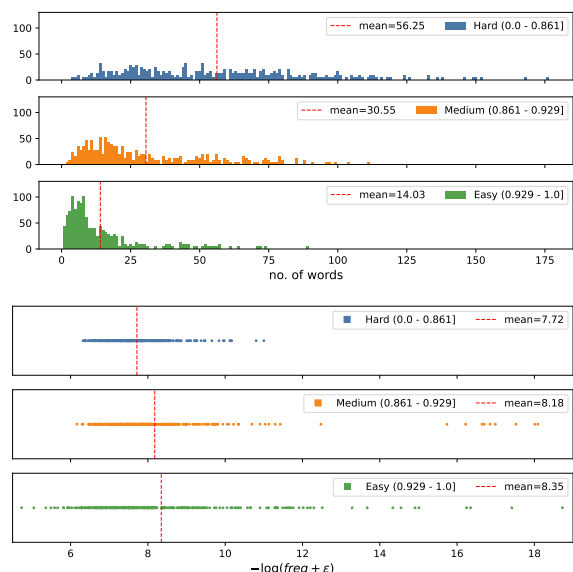


Figure 2: (Top) Sentence length distribution per bin. (Bottom) Rarity score distribution per bin.

4 Models

4.1 TowerMistral 7B

TowerMistral 7B (Alves et al., 2024) is an LLM specifically fine-tuned for MT. It is built on the Mistral (Chaplot, 2023) foundation model and belongs to the “Tower” family of models developed by Unbabel. These models are designed to excel at multilingual translation by leveraging the strong latent representation and reasoning capabilities of their base models. The Tower architecture’s key innovation is its tailored cross-lingual training objective and careful fine-tuning on high-quality parallel corpora. This enables Tower models to achieve state-of-the-art performance on neural MT benchmarks like WMT24 (Kocmi et al., 2024a).

4.2 Gemma-3

Gemma-3 (Team et al., 2025) is a family of lightweight, state-of-the-art openly available models from Google. For our experiments, we utilize the instruction tuned 4-billion parameter variant, google/gemma-3-4b-it. While not specifically designed for translation like the Tower models, its strong general language understanding and generation capabilities, built from recent advancements in Google’s Gemini research, make it a compelling baseline for evaluating the performance of LLMs on MT in a few-shot or zero-shot setting. This allows us to compare a specialized translation model against a powerful, general-knowledge counterpart.

4.3 Quantized Models

We used quantized versions of both the TowerMistral and Gemma models, which were published by TensorBlock on Hugging Face. TensorBlock’s implementation uses mixed-precision, weight-only Post-Training Quantization (PTQ). This approach quantizes only the model’s weights, leaving activations and the KV cache in higher precision, and does not involve any retraining after the quantization process. Mixed-precision means if the model is quantized to k bits, some weights are quantized to k bits, whilst other weights are quantized to $> k$ bits in an effort to preserve quality. Some weights are not quantized at all.

5 Results

5.1 Translation Quality Metrics

We evaluate translation quality primarily using COMET-22 (Rei et al., 2022a), a reference-based

metric designed to assess semantic adequacy and fluency. For completeness, we also computed chrF (Popović, 2015) and COMETKiwi-XL (Rei et al., 2022b), a character-level overlap metric and a reference-free variant of COMET, respectively. Detailed results for the additional metrics are provided in Appendix A, as well as a small subset of sentences representing qualitative results in Appendix B.

5.2 Gemma-3 Comet Results

As shown in Figure 3, the plot presents the difference in COMET scores (Δ COMET vs G3) between quantized Gemma-3 models and the full-precision baseline across language pairs and difficulty levels. The full-precision baseline (G3) and its 8- and 4-bit quantized versions performed nearly identically across all levels (Easy, Medium, Hard) and language pairs, indicating minimal impact of quantization. In contrast, the G3_2bit model exhibited notable degradation, with Δ COMET increasing as task difficulty increased: Easy (-0.06 to -0.10), Medium (-0.11 to -0.16) and Hard (-0.16 to -0.20). This shows that extreme quantization begins to impair the model’s semantic fidelity and fluency.

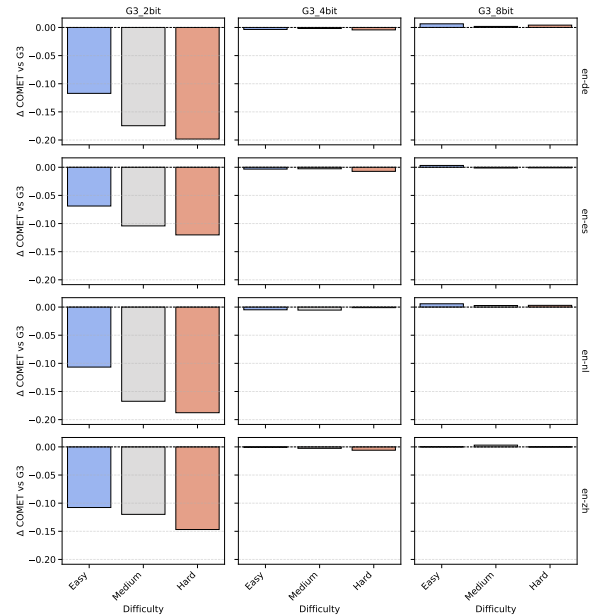


Figure 3: Δ COMET Scores (Baseline - G3) by Difficulty, Model Variant, and Language Pair.

5.3 TowerMistral Comet Results

As shown in Figure 4, quantization had negligible impact on COMET scores across all difficulty levels and models. Even the most aggressively quan-

tized 2-bit model achieved scores that were nearly indistinguishable from the full-precision baseline. COMET scores remained consistently high across Easy, Medium, and Hard subsets (Figure 5), with only marginal variations that fall well within expected noise levels.

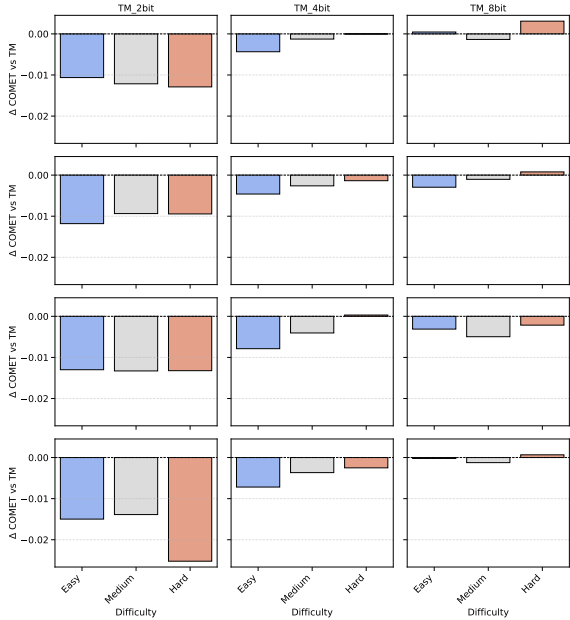


Figure 4: Δ COMET Scores (Baseline - TM) by Difficulty, Model Variant, and Language Pair.

5.4 Comparison: TowerMistral vs. Gemma-3

Figure 5 shows the COMET score differences for both TM and G3 across four quantization levels, broken down by difficulty tiers and specific language pairs. The most striking observation is the contrast in robustness between the two models. TM maintains consistent and high COMET scores, generally clustering between 0.86 and 0.77, with its performance remaining nearly flat even at the most aggressive 2-bit quantization.

In contrast, G3 exhibits substantial performance decreases as quantization increases, especially at the 2-bit level. While G3’s full-precision scores are initially comparable to TM’s, the model’s performance plummets dramatically at 2-bit, with the Hard subset dropping to approximately 0.55. Consequently, TM consistently outperforms G3, with the performance gap becoming particularly significant at 2-bit. This resilience means that TM’s 2-bit performance (e.g., Easy ≈ 0.86) sometimes matches or even surpasses G3’s full-precision baseline (G3 Easy ≈ 0.85), strongly underscoring TM’s ability to maintain translation quality under com-

pression.

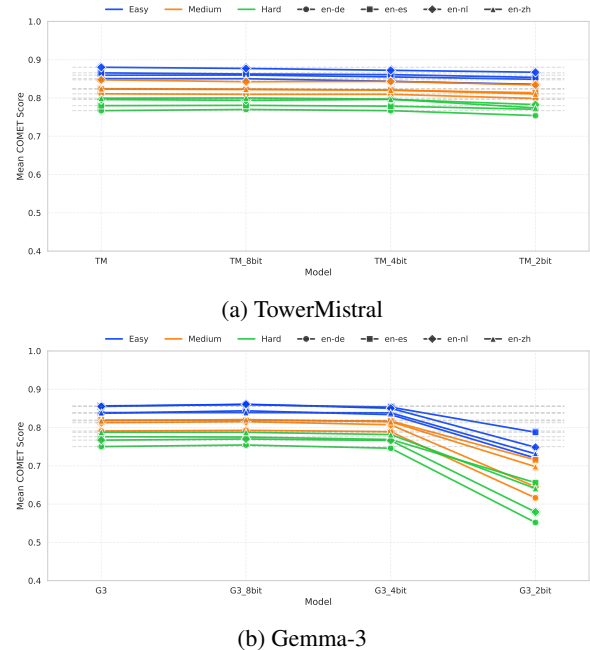


Figure 5: Mean COMET scores by difficulty, model variant, and language pair.

6 Conclusion

Our analysis reveals that quantization affects translation performance in nuanced ways depending on model architecture and sentence difficulty. While both TowerMistral and Gemma-3 maintain high semantic quality at moderate quantization (8- and 4-bit), differences emerge sharply at 2-bit precision. Gemma-3 experiences marked degradation, especially for Easy and Hard inputs, whereas TowerMistral preserves COMET scores near baseline across all difficulty tiers and languages.

These results indicate that TowerMistral’s architecture is inherently more resilient to precision loss, sustaining semantic fidelity and fluency even under extreme compression. More broadly, we find that quantization disproportionately impacts very short and syntactically complex sentences, suggesting that sentence-level difficulty modulates robustness to quantization noise. Overall, TowerMistral achieves a more favorable balance between efficiency and translation quality, making it a stronger candidate for low-resource or latency-sensitive machine translation deployments.

7 Limitations & Future Work

Our study relies on automatically estimated sentence difficulty scores, which predicts difficulty using features trained on ESA data. While this approach provides a practical proxy for difficulty segmentation, it may not perfectly capture the true cognitive or linguistic complexity of each source sentence. Additionally, our experiments were conducted using pre-quantized model variants, limiting fine-grained control over quantization dynamics and calibration. While different language pairs are analyzed, they all involve high-resource languages, restricting insights into the impact on low-resource languages.

Future work will explore end-to-end quantization procedures to better understand how training-time quantization strategies influence translation robustness across difficulty levels. We also plan to validate difficulty estimation with human-annotated data and extend the analysis to a wider variety of languages, model families, and adaptive quantization methods that adjust precision based on input complexity.

References

- Sweta Agrawal and Marine Carpuat. 2019. Controlling text complexity in neural machine translation.
- Alham Fikri Aji and Kenneth Heafield. 2019. Compressing neural machine translation models with 4-bit precision.
- Duarte M Alves, José Pombal, Nuno M Guerreiro, Pedro H Martins, João Alves, Amin Farajian, Ben Peters, Ricardo Rei, Patrick Fernandes, Sweta Agrawal, et al. 2024. Tower: An open multilingual large language model for translation-related tasks. *arXiv preprint arXiv:2402.17733*.
- Duygu Ataman, Alexandra Birch, Nizar Habash, Marcello Federico, Philipp Koehn, and Kyunghyun Cho. 2025. [Machine translation in the era of large language models: a survey of historical and emerging problems](#). *Information*, 16(9).
- Aishwarya Bhandare, Vamsi Sripathi, Deepthi Karkada, Vivek Menon, Sun Choi, Kushal Datta, and Vikram Saletore. 2019. Efficient 8-bit quantization of transformer neural machine language translation model.
- Devendra Singh Chaplot. 2023. Albert q. jiang, alexandre sablayrolles, arthur mensch, chris bamford, devendra singh chaplot, diego de las casas, florian bressand, gianna lengyel, guillaume lample, lucile saulnier, l  lio renard lavaud, marie-anne lachaux, pierre stock, teven le scao, thibaut lavril, thomas wang, timoth  e lacroix, william el sayed. *arXiv preprint arXiv:2310.06825*, 3.
- Insoo Chung, Byeongwook Kim, Yoonjung Choi, Se Jung Kwon, Yongkweon Jeon, Baeseong Park, Sangha Kim, and Dongsoo Lee. 2020. Extremely low bit transformer quantization for on-device neural machine translation.
- Daniel Deutsch, Eleftheria Briakou, Isaac Caswell, Mara Finkelstein, Rebecca Galor, Juraj Juraska, Geza Kovacs, Alison Lui, Ricardo Rei, Jason Riesa, Shruti Rijhwani, Parker Riley, Elizabeth Salesky, Firas Tribelsi, Stephanie Winkler, Biao Zhang, and Markus Freitag. 2025. [WMT24++: Expanding the Language Coverage of WMT24 to 55 Languages Dialects](#).
- Tom Kocmi, Eleftherios Avramidis, Rachel Bawden, Ondrej Bojar, Anton Dvorkovich, Christian Federmann, Mark Fishel, Markus Freitag, Thamme Gowda, Roman Grundkiewicz, et al. 2024a. Preliminary wmt24 ranking of general mt systems and llms. *arXiv preprint arXiv:2407.19884*.
- Tom Kocmi, Vil  m Zouhar, Eleftherios Avramidis, Roman Grundkiewicz, Marzena Karpinska, Maja Popovi  , Mrinmaya Sachan, and Mariya Shmatova. 2024b. [Error span annotation: A balanced approach for human evaluation of machine translation](#). In *Proceedings of the Ninth Conference on Machine Translation*, pages 1440–1453, Miami, Florida, USA. Association for Computational Linguistics.
- Zuchao Li, Rui Wang, Kehai Chen, Masao Utiyama, Eiichiro Sumita, Zhuosheng Zhang, and Hai Zhao. 2020. [Explicit sentence compression for neural machine translation](#). In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 8238–8245.
- Alireza Mohammadshahi, Vassilina Nikoulina, Alexandre Berard, Caroline Brun, James Henderson, and Laurent Besacier. 2022. What do compressed multilingual machine translation models forget?
- OpenAI. 2024. [Gpt-4 technical report](#).
- Maja Popovi  . 2015. [chrF: character n-gram F-score for automatic MT evaluation](#). In *Proceedings of the Tenth Workshop on Statistical Machine Translation*, pages 392–395, Lisbon, Portugal. Association for Computational Linguistics.
- Lorenzo Proietti, Stefano Perrella, Vil  m Zouhar, Roberto Navigli, and Tom Kocmi. 2025. [Estimating machine translation difficulty](#).
- Ricardo Rei, Jos   G. C. de Souza, Duarte Alves, Chrysoula Zerva, Ana C Farinha, Taisiya Glushkova, Alon Lavie, Luisa Coheur, and Andr   F. T. Martins. 2022a. [COMET-22: Unbabel-IST 2022 submission for the metrics shared task](#). In *Proceedings of the Seventh Conference on Machine Translation (WMT)*, pages 578–585, Abu Dhabi, United Arab Emirates (Hybrid). Association for Computational Linguistics.
- Ricardo Rei, Marcos Treviso, Nuno M. Guerreiro, Chrysoula Zerva, Ana C Farinha, Christine Maroti, Jos   G. C. de Souza, Taisiya Glushkova, Duarte

- Alves, Luisa Coheur, Alon Lavie, and André F. T. Martins. 2022b. [CometKiwi: IST-unbabel 2022 submission for the quality estimation shared task](#). In *Proceedings of the Seventh Conference on Machine Translation (WMT)*, pages 634–645, Abu Dhabi, United Arab Emirates (Hybrid). Association for Computational Linguistics.
- Gemma Team, Aishwarya Kamath, Johan Ferret, Shreya Pathak, Nino Vieillard, Ramona Merhej, Sarah Perrin, Tatiana Matejovicova, Alexandre Ramé, Morgane Rivi re, et al. 2025. Gemma 3 technical report. *arXiv preprint arXiv:2503.19786*.
- Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timoth e Lacroix, Baptiste Rozi re, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. 2023. [Llama: Open and efficient foundation language models](#).
- Yu Wan, Baosong Yang, Derek Fai Wong, Lidia Sam Chao, Liang Yao, Haibo Zhang, and Boxing Chen. 2022. [Challenges of neural machine translation for short texts](#). *Computational Linguistics*, 48(2):321–342.
- Wayne Xin Zhao, Kun Zhou, Junyi Li, Tianyi Tang, Xiaolei Wang, Yupeng Hou, Yingqian Min, Beichen Zhang, Junjie Zhang, Zican Dong, Yifan Du, Chen Yang, Yushuo Chen, Zhipeng Chen, Jinhao Jiang, Ruiyang Ren, Yifan Li, Xinyu Tang, Zikang Liu, Peiyu Liu, Jian-Yun Nie, and Ji-Rong Wen. 2025. [A survey of large language models](#).
- Xunyu Zhu, Jian Li, Yong Liu, Can Ma, and Weiping Wang. 2024. [A survey on model compression for large language models](#).

A Additional Quantitative Results

This appendix presents additional quantitative results comparing TowerMistral (TM) and Gemma-3 (G3) across quantization levels and difficulty tiers. Each plot shows mean scores by model and language pair.

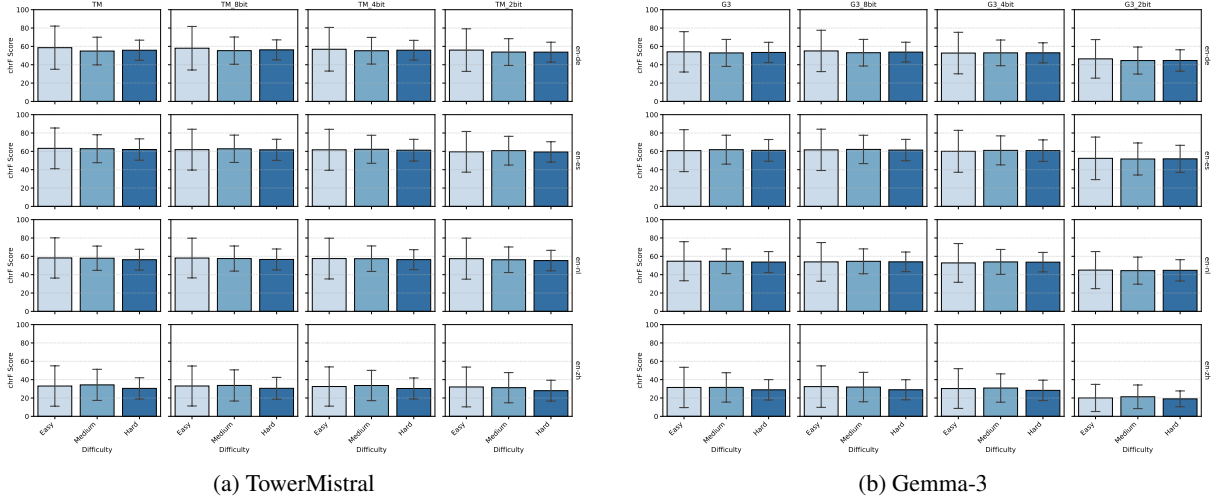


Figure 6: Mean chrF scores by difficulty, model variant, and language pair.

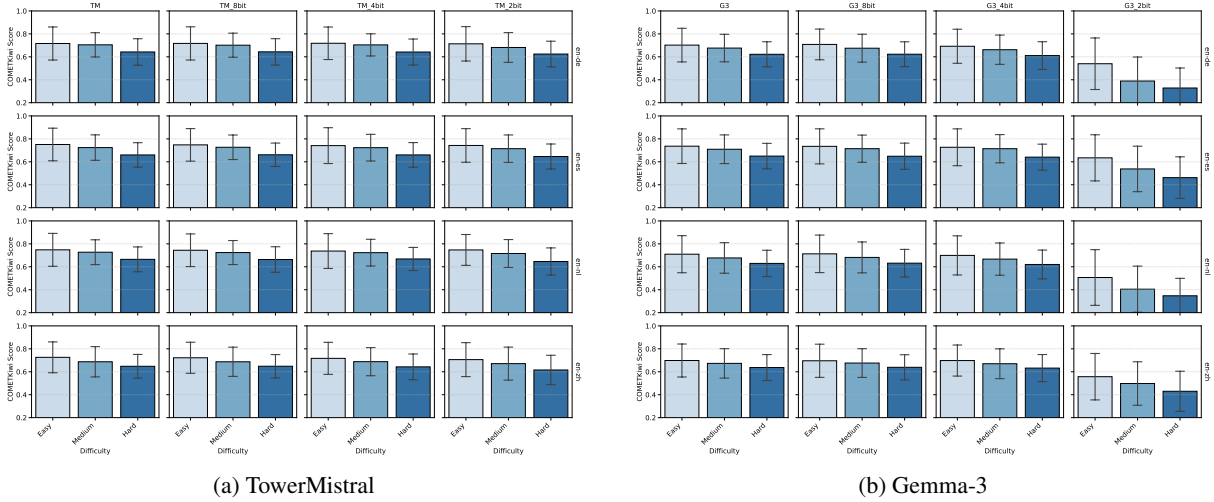


Figure 7: Mean COMETKiwi-XL scores by difficulty, model variant, and language pair.

B Additional Qualitative Results

This appendix provides qualitative examples of translations from TowerMistral and Gemma-3 across different quantization levels.

Model	Prediction (German)	Backtranslation (English)
TowerMistral 7B		
Source:	<i>The pictures are from Saturday. We had a flat tire during our last landing.</i>	
Baseline	Die Bilder sind vom Samstag. Nach der letzten Landung war ein Reifen platt.	Pics are from Saturday, ended up with a flat tire after our last landing.
8-bit	Die Bilder sind vom Samstag, bei unserer letzten Landung hatten wir einen Platten.	The pictures are from Saturday. We had a flat tire during our last landing.
4-bit	Die Bilder sind vom Samstag, bei unserer letzten Landung hatten wir einen Platten.	The pictures are from Saturday. We had a flat tire during our last landing.
2-bit	Die Bilder sind vom Samstag, und wir hatten einen Reifenschaden nach unserer letzten Landung.	The pictures are from Saturday, and we had a damaged tire after our last landing.
Gemma-3		
Source:	<i>I am genuinely sorry to read that. :(</i>	
Baseline	Es tut mir wirklich leid, das zu lesen :(	I am genuinely sorry to read that. :(
8-bit	Es tut mir wirklich leid, das zu lesen :(	I am genuinely sorry to read that. :(
4-bit	Es tut mir wirklich leid, das zu lesen :(	I am genuinely sorry to read that. :(
2-bit	Ich bin aufrichtet, dass ich lese, dass. :(	I am genuine, that I read, that. :(

Table 1: German translations from TowerMistral 7B and Gemma-3 across quantization levels. The source sentence is shown above each set of predictions, with corresponding English backtranslations.

Model	Prediction (Dutch)	Backtranslation (English)
TowerMistral 7B		
Source:	<i>Lovelies and gentle lovelies,</i>	
Baseline	Liefjes en zachte liefjes,	Lovelies and gentle lovelies,
8-bit	Lieve mensen,	Sweet people,
4-bit	Lieve mensen,	Sweet people,
2-bit	Lieve schatjes en lieve, lieve schatjes,	Dear darlings and dear, dear darlings,
Gemma-3		
Source:	<i>I'm a solid proponent of using protective gear!</i>	
Baseline	Ik ben een sterke voorstander van het gebruik van beschermende kleding!	I am a strong advocate of using protective clothing!
8-bit	Ik ben een sterke voorstander van het gebruik van beschermingsmiddelen!	I am a strong advocate of the use of protective equipment!
4-bit	Ik ben een sterke voorstander van het dragen van beschermende uitrusting!	I am a strong advocate of wearing protective gear!
2-bit	Ik ben een stevige pleger van het gebruiken van beschermend materiaal!	I am a firm believer in using protective equipment!

Table 2: Dutch translations from TowerMistral 7B and Gemma-3 across quantization levels. The source sentence is shown above each set of predictions, with corresponding English backtranslations.